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ABSTRACT

The **Automated Book Picking Robot** is an intelligent robotic system developed to automate the process of locating, identifying, and retrieving books from library racks with minimal human intervention. Libraries are essential knowledge centres in educational institutions, but traditional library operations largely depend on manual effort for searching, issuing, and organizing books. As the number of books and users increases, manual management becomes time-consuming, inefficient, and prone to errors. The proposed system aims to overcome these challenges by integrating robotics, embedded systems, RFID technology, and automated control mechanisms into a single efficient solution.

The core objective of this project is to design and implement a robotic system capable of automatically navigating to a specified rack location, identifying a selected book using RFID technology, and picking it using a servo-based robotic arm mechanism. The system is controlled by the powerful **Arduino MEGA 2560 microcontroller**, which acts as the central processing unit for coordinating sensors, motors, and input-output modules. The use of Arduino Mega provides sufficient digital and analogue pins to handle multiple components such as servo motors, line follower sensors, RFID reader, LCD display, and motor driver without conflicts.

In the proposed system, each book in the library is attached with an RFID tag that contains a unique identification number (UID). When a user selects a book using the LCD interface and control buttons, the robot retrieves the corresponding rack and shelf data from its programmed database. The robot then moves along a predefined path using line follower sensors, ensuring accurate navigation within the library layout. Upon reaching the correct rack, the robotic arm adjusts its height and orientation using multiple servo motors to match the shelf position. The RFID reader scans the book tags to confirm the correct book before gripping it securely. After successful retrieval, the robot returns to its starting position.

The system integrates several hardware components including DC motors for movement, L298N motor driver for motor control, servo motors for robotic arm movement, RFID module (RC522) for identification, and a regulated power supply system to ensure stable operation. The software is developed using the Arduino IDE, where embedded C programming is used to implement navigation logic, identification algorithms, and arm control sequences. The entire process is controlled through a well-defined algorithm that ensures reliability, accuracy, and repeatability.

This project demonstrates the practical application of automation in library management systems. It reduces the time required to search and retrieve books, minimizes human effort, and decreases the chances of misplacement or identification errors. Additionally, the system enhances operational efficiency and improves user experience, especially in large academic libraries where manual searching can be challenging.

The Automated Book Picking Robot also serves as a foundation for future advancements in smart library systems. The system can be further enhanced by integrating Internet of Things (IoT) connectivity, cloud-based databases, and artificial intelligence techniques such as deep learning for visual book recognition and autonomous navigation. Such improvements would transform traditional libraries into fully automated smart knowledge centres.

In conclusion, the Automated Book Picking Robot is an innovative and practical solution that addresses key challenges in library management. By combining robotics, embedded systems, and RFID technology, the system provides an efficient, scalable, and reliable approach to automated book retrieval. This project not only improves library operations but also highlights the growing importance of automation in educational institutions and modern smart environments.

CHAPTER 1 – INTRODUCTION

1.1 Overview

The **Automated Book Picking Robot** is an intelligent robotic system designed to automate the process of locating, identifying, and retrieving books from library shelves. In traditional libraries, students and staff manually search for books by moving through racks and scanning shelves. This process consumes valuable time, increases physical effort, and sometimes leads to errors such as book misplacement or difficulty in locating the required material. The proposed system introduces a smart automation solution that integrates robotics, embedded systems, and RFID technology to improve efficiency and accuracy in library operations.

The system is primarily controlled by the **Arduino MEGA 2560 microcontroller**, which serves as the central processing unit of the robot. It coordinates various hardware components including sensors, motors, the RFID reader, and the LCD display module. The Arduino Mega is selected because it offers a large number of input and output pins, making it suitable for projects involving multiple servo motors and sensors without pin conflicts. The programming logic for navigation, identification, and arm movement is written using embedded C in the Arduino IDE.

The working of the system begins when a user selects a desired book using a 16×2 LCD display and push-button interface. The selected book details, including rack and shelf position, are stored in the robot's internal database. Once the selection is confirmed, the robot begins navigation. It follows a predefined path using line follower sensors that detect a black line on the floor. This ensures accurate and repeatable movement toward the correct rack location.

Each book in the library is equipped with an RFID tag containing a unique identification number (UID). After reaching the designated rack, the robot activates the RFID reader module to scan and verify the correct book. The identification process ensures that the robot retrieves the exact book selected by the user, reducing errors and improving reliability.

The robotic arm mechanism plays a crucial role in the system. It consists of multiple servo motors responsible for base rotation, vertical lifting, elbow movement, wrist adjustment, and gripping action. These servo motors allow precise positioning and careful

handling of books. Once the book is securely gripped, the robot returns to its starting point, completing the retrieval process.

The movement of the robot is controlled using DC motors driven by the L298N motor driver module. The motor driver enables forward and backward motion and ensures sufficient torque to carry the robotic arm and book load. A regulated power supply unit provides stable voltage to all components, ensuring smooth operation without fluctuations.

The overview of this project highlights the integration of mechanical design, electronic components, and software programming to create a complete automated system. It demonstrates the practical application of robotics in real-world environments, particularly in educational institutions. By automating the book retrieval process, the system reduces manual effort, improves speed, enhances accuracy, and supports the transformation of traditional libraries into smart libraries.

In summary, the Automated Book Picking Robot is a cost-effective and scalable solution for library automation. It combines embedded systems, sensor technology, motor control, and robotic arm mechanisms into a unified system capable of performing reliable and efficient book retrieval. This project not only improves library management but also showcases the innovative use of robotics in solving practical challenges in educational settings.

1.2 Aim and Objectives

Aim

The primary aim of the **Automated Book Picking Robot** project is to design and develop an intelligent robotic system that can automatically locate, identify, and retrieve books from library racks using RFID technology, embedded systems, and a servo-based robotic arm mechanism. The system is intended to reduce manual effort, minimize search time, improve accuracy in book retrieval, and enhance overall efficiency in library management.

The project aims to create a practical and cost-effective automation solution that can be implemented in educational institutions such as colleges and universities. By integrating hardware components and software control algorithms, the system demonstrates how robotics can be applied in real-world environments to solve operational challenges in libraries. The long-term goal is to contribute toward transforming traditional libraries into smart, automated knowledge centres.

Objectives

To achieve the above aim, the following objectives are defined:

2. **To Design an Automated Navigation System**

The robot must be capable of moving autonomously inside the library environment. This is achieved using line follower sensors that detect a predefined path on the floor. The navigation system should ensure accurate movement toward specific rack locations without human guidance.

2. To Implement RFID-Based Book Identification

Each book in the library is equipped with an RFID tag containing a unique identification number (UID). The system must scan and verify the selected book using an RFID reader module to ensure correct identification and reduce errors during retrieval.

3. To Develop a User-Friendly Interface

The project includes a 16×2 LCD display and control buttons that allow users to scroll through available book options and select the required book. The interface should be simple, clear, and easy to operate.

4. To Design a Multi-Servo Robotic Arm

The robotic arm must be capable of:

- Adjusting to different shelf heights
- Rotating toward the required position
- Gripping the selected book securely
- Handling the book carefully without damage

This is achieved using multiple servo motors for base rotation, lifting, and gripping mechanisms.

5. To Ensure Stable Motor Control

The DC motors responsible for robot movement must be controlled using the L298N motor driver module. The system should provide smooth forward and backward movement and maintain balance while carrying the book.

6. To Implement an Efficient Control Algorithm

A structured algorithm must be developed to manage the complete process, including:

- Book selection
- Path navigation
- Rack detection
- RFID scanning
- Arm movement
- Return to starting position

The algorithm should ensure accuracy, reliability, and repeatability.

7. To Provide Proper Power Management

The system must include a regulated power supply to maintain stable voltage levels for the microcontroller, sensors, and motors. Proper grounding and voltage regulation are essential for safe and stable operation.

8. To Reduce Time and Human Effort

One of the key objectives is to significantly reduce the time required to search for books and decrease the workload of library staff.

9. To Create a Scalable and Upgradable System

The project should be designed in a way that allows future enhancements such as IoT connectivity, mobile application control, and AI-based automation.

1.3 Scope of the Project

The **Automated Book Picking Robot** project is developed to introduce automation into traditional library systems by implementing a robotic mechanism capable of locating and retrieving books efficiently. The scope of this project defines the boundaries, applications, and potential expansion areas of the system. It explains where the system can be applied, what it currently covers, and how it can be enhanced in the future.

Present Scope of the Project

The current implementation of the Automated Book Picking Robot is designed primarily for **small to medium-sized academic libraries** where books are arranged systematically in racks and shelves. The robot operates in a structured indoor environment where a predefined path is marked for navigation.

The present system includes the following capabilities:

- **Book Selection Interface:**
  Users can select a book using an LCD display and push buttons.
- **Predefined Rack Mapping:**
  The rack number and shelf height are pre-programmed into the system database.
- **Line-Based Navigation:**
  The robot follows a black line path using line follower sensors to reach the correct rack.
- **RFID-Based Book Identification:**
  Each book contains an RFID tag, and the system verifies the correct book before picking it.
- **Servo-Based Robotic Arm:**
  The robotic arm adjusts to different shelf heights and grips the book securely.
- **Return to Starting Position:**
  After retrieving the book, the robot returns to its original location.

The current scope mainly focuses on demonstrating a **functional prototype model** that integrates hardware components such as Arduino MEGA 2560, RFID module, motor driver, servo motors, and sensors to perform automated book retrieval.

Technical Scope

From a technical perspective, the project covers multiple domains of engineering and computer science, including:

- Embedded systems programming
- Robotics and automation
- Sensor integration
- Motor control systems
- Power regulation and hardware interfacing
- Algorithm development

This project provides practical exposure to real-time control systems and demonstrates how hardware and software can work together in an automated environment.

Operational Scope

The robot is designed for structured and organized libraries where:

- Racks are placed in fixed positions.
- Shelf heights are predetermined.
- A path for navigation is clearly defined.
- Books are arranged in systematic order.

It does not currently support dynamic rearrangement of racks or unstructured environments. The system operates based on pre-programmed mapping rather than real-time visual recognition.

Limitations Within Current Scope

The current version of the system:

- Works only in controlled indoor environments.
- Requires predefined path markings.
- Depends on RFID tags for identification.
- Does not include obstacle detection or AI-based navigation.

These limitations define the boundaries of the present scope and help identify future improvement areas.

Future Scope of the Project

The Automated Book Picking Robot has significant potential for expansion and enhancement. Future improvements may include:

1. **IoT Integration:**
   Connecting the system to a cloud database for real-time inventory tracking.
2. **Mobile Application Control:**
   Allowing users to request books through a smartphone application.
3. **Deep Learning and Computer Vision:**
   Implementing camera-based book recognition and autonomous navigation.
4. **Obstacle Detection System:**
   Adding ultrasonic sensors for dynamic obstacle avoidance.
5. **Multi-Robot Coordination:**
   Deploying multiple robots in large libraries for faster operations.
6. **Autonomous Shelf Scanning:**
   Automatically detecting misplaced books.
7. **Integration with Digital Library Systems:**
   Synchronizing with existing library management software.

Application Scope Beyond Libraries

Although designed for libraries, the system concept can be extended to:

- Warehouse automation
- Archive management systems
- Industrial storage units
- Document retrieval systems
- Inventory management in retail stores

The robotic retrieval mechanism can be adapted for handling other lightweight objects in structured environments.

1.4 Problem Statement

Libraries are essential centres of learning in educational institutions, providing students and faculty with access to textbooks, reference materials, journals, and research documents. However, despite the availability of digital catalos systems, the physical process of locating and retrieving books from racks remains largely manual. As the number of books and users increases, traditional library systems face several operational challenges that reduce efficiency and productivity. These challenges form the core problem addressed by the Automated Book Picking Robot project.

One of the primary problems in conventional libraries is the **time-consuming manual search process**. Even when a student knows the book title or call number, locating the exact rack and shelf can take considerable time. In large college libraries with thousands of books arranged across multiple racks, searching manually becomes inefficient, especially during peak academic hours. This leads to delays and reduces the overall effectiveness of the learning environment.

Another significant issue is the **misplacement of books**. Books are often returned to incorrect shelves either accidentally or due to lack of proper monitoring. A single misplaced book can be extremely difficult to locate later, even if it exists in the library inventory system. This creates confusion for students and additional workload for library staff.

The **high workload on librarians** is also a major concern. Librarians are responsible for issuing and returning books, maintaining records, assisting students, organizing shelves, and managing inventory. When many students require assistance simultaneously, the staff may become overburdened. This reduces service quality and increases the chance of human errors.

Human error is another important problem in manual systems. Mistakes in recording transactions, incorrect shelving, or miscommunication can lead to inaccurate inventory data. Over time, such errors accumulate and affect the reliability of the library management system.

Additionally, traditional libraries lack **real-time tracking and automation**. Although digital catalos systems may indicate whether a book is available, they do not guarantee its exact physical location. Manual verification is often required, which consumes time and effort. There is no automated mechanism to ensure that books are stored in the correct position.

Another issue is **limited accessibility**. Large libraries can be difficult to navigate for elderly users or differently-abled students. Walking through multiple racks to search for books may not be convenient for all users. An automated retrieval system would improve accessibility and make library services more inclusive.

Security concerns also contribute to the problem. Book theft or unauthorized removal can occur in large libraries. Without automated monitoring systems, it is difficult to track and prevent such incidents effectively.

Considering these challenges, the core problem can be summarized as follows:

Traditional library systems rely heavily on manual book searching and retrieval processes, which are time-consuming, error-prone, inefficient, and labour-intensive. There is a need for an automated system that can accurately locate, identify, and retrieve books with minimal human intervention.

The Automated Book Picking Robot is proposed as a solution to address these problems. By integrating RFID technology, robotic navigation, servo-based arm mechanisms, and embedded control systems, the project aims to reduce search time, minimize human errors, improve efficiency, and enhance overall library management.

1.5 Need for Automation in Libraries

Libraries are one of the most important academic resources in educational institutions. They store thousands of books, journals, research papers, and reference materials that support learning and knowledge development. However, many libraries still operate using traditional manual methods for searching, issuing, returning, and organizing books. As the number of students and books increases, manual systems become inefficient and difficult to manage. Therefore, there is a strong need for automation in libraries to improve efficiency, accuracy, and service quality.

One of the main reasons for automation is **reducing search time**. In a traditional library, students must physically walk through racks and shelves to locate a book. Even if a digital catalo shows that the book is available, finding its exact location can take several minutes or even longer. During busy academic periods, this leads to crowding and delays. Automation systems, such as robotic book retrieval mechanisms, can quickly identify and bring the required book, saving valuable time for both students and staff.

Another important need is **minimizing human errors**. Manual systems are prone to mistakes such as incorrect shelving, inaccurate record entry, or misplacement of books. A single misplaced book can become difficult to locate, even though it exists in the system database. Automated systems using technologies like RFID ensure accurate identification and tracking of each book. This reduces errors and improves reliability in library operations.

Automation also helps in **reducing workload on librarians**. Librarians are responsible for multiple tasks including managing inventory, assisting students, updating records, and organizing shelves. Handling all these activities manually can be exhausting and time-consuming. By automating repetitive tasks such as book searching and retrieval, staff can focus on more meaningful responsibilities such as academic assistance and resource planning.

Another major reason for automation is **real-time inventory tracking**. In manual systems, checking the availability of books requires physical verification. Automated systems maintain digital records that update instantly whenever a book is issued or returned. This ensures transparency and accurate stock management.

Automation improves **security and monitoring** in libraries. Book theft or unauthorized removal can cause losses. RFID-based systems can detect books passing through exit points without proper authorization. This adds an additional layer of security and protects valuable resources.

Accessibility is another key factor. Large libraries may be difficult to navigate for elderly users or differently-abled students. Automated retrieval systems allow users to request books without physically moving across racks. This makes the library more inclusive and user-friendly.

With the advancement of technology, educational institutions are transforming into **smart campuses**. Classrooms, attendance systems, and administrative processes are becoming digital. Libraries must also evolve to match this technological progress. Automation ensures that libraries remain modern, efficient, and aligned with smart education initiatives.

Furthermore, automation supports **scalability and future growth**. As institutions expand, the number of books and users increases. Manual systems struggle to handle large-scale operations efficiently. Automated systems can manage higher volumes with minimal additional manpower, making them cost-effective in the long term.

In the context of the Automated Book Picking Robot project, automation directly addresses the problem of manual book retrieval. The robot uses sensors, RFID technology, and a robotic arm to locate and pick books automatically. This demonstrates how robotics can be practically applied in library environments to enhance service efficiency.

CHAPTER 2 – SYSTEM ANALYSIS

2.1 Literature Survey / Related Work

The development of the **Automated Book Picking Robot** is based on concepts and technologies that have been explored in the fields of robotics, embedded systems, RFID-based identification, and library automation. A literature survey helps in understanding existing research contributions, identifying technological advancements, and analysing how previous systems have addressed similar problems. This section reviews related works and highlights the evolution of automation in libraries and robotic retrieval systems.

Library Automation Systems

Several researchers have proposed RFID-based library management systems to improve book tracking and inventory management. RFID technology has been widely adopted in modern libraries for issuing, returning, and monitoring books. Unlike barcode systems, RFID does not require direct line-of-sight scanning and allows faster processing of multiple books simultaneously. Studies have shown that RFID improves accuracy, reduces human effort, and enhances security in libraries.

However, most RFID-based systems focus only on book identification and transaction management. They do not automate the physical process of retrieving books from shelves. The Automated Book Picking Robot extends this concept by combining RFID identification with robotic movement and mechanical retrieval.

Robotic Navigation Systems

Autonomous mobile robots have been extensively studied for indoor navigation. Line-following robots are among the most common beginner-level robotic systems used for structured path navigation. These robots use infrared sensors to detect black lines on white surfaces and move accordingly. Research in this area has demonstrated that line-following systems are reliable and cost-effective for controlled environments.

More advanced navigation systems use obstacle detection sensors, SLAM (Simultaneous Localization and Mapping), and AI-based algorithms. However, for structured environments such as libraries with fixed rack arrangements, line-following navigation provides sufficient accuracy while keeping the system simple and affordable.

Robotic Arm Mechanisms

Robotic arms are widely used in industrial automation for object handling, assembly, and material transportation. Research in robotics has focused on improving precision, grip strength, and movement flexibility using servo motors and stepper motors. Multi-degree-of-freedom robotic arms are capable of performing complex movements such as lifting, rotating, and gripping.

In the context of book retrieval, servo motor-based arms are suitable because they provide accurate angular positioning and are easy to control using microcontrollers. Previous studies in pick-and-place robots have demonstrated that servo-controlled arms are effective for handling lightweight objects with precision.

Embedded Systems in Automation

Microcontrollers such as Arduino boards are widely used in automation projects due to their simplicity, affordability, and flexibility. Many research projects in robotics and automation rely on Arduino platforms for sensor integration and motor control. The Arduino MEGA 2560, in particular, is preferred for projects requiring multiple input-output connections.

Embedded system research emphasizes real-time processing, low power consumption, and efficient control algorithms. The Automated Book Picking Robot utilizes embedded programming principles to integrate sensors, motors, and identification modules into a synchronized system.

Smart Library Concepts

Recent advancements in smart library systems include IoT integration, cloud-based management, and AI-powered recommendation systems. Research papers have proposed intelligent shelves that automatically detect book positions and generate alerts for misplaced books. Some experimental models have explored robotic assistants capable of guiding users to book locations.

However, fully autonomous book retrieval robots are still under development and not widely implemented due to cost and complexity factors. The proposed project provides a prototype-level solution that demonstrates the feasibility of robotic book picking in a controlled environment.

Research Gap Identified

From the reviewed literature, it is evident that:

- RFID is widely used for identification but not commonly integrated with robotic retrieval.
- Robotic navigation systems are well-studied but rarely applied specifically in library automation.
- Most library automation systems focus on digital management rather than physical book handling.

The Automated Book Picking Robot bridges this gap by integrating RFID-based identification, robotic navigation, and servo-based object handling into a single system.

2.2 Proposed System Overview

The **Automated Book Picking Robot** is a smart robotic system designed to automate the physical process of locating and retrieving books from library racks. The proposed system integrates embedded systems, RFID technology, line-following navigation, and a servo-based robotic arm to create a reliable and efficient automated solution for library management. Unlike traditional systems that rely solely on digital catalogues for tracking, this system physically retrieves the selected book without manual assistance.

The central controller of the proposed system is the Arduino MEGA 2560. This microcontroller acts as the brain of the robot, coordinating all hardware components such as sensors, motors, the RFID reader, LCD display, and control buttons. The Arduino Mega is chosen because of its large number of digital and analogy input/output pins, which are essential for handling multiple servo motors and sensors simultaneously without pin conflicts.

System Architecture Overview

proposed system consists of the following major subsystems:

1. **User Interface Subsystem**
2. **Navigation Subsystem**
3. **Identification Subsystem**
4. **Robotic Arm Mechanism**
5. **Motor Control Subsystem**
6. **Power Management Unit**

Each subsystem works together to complete the automated book retrieval process.

1. User Interface Subsystem

The system includes a 16×2 LCD display and push-button controls. The LCD displays the list of available books stored in the robot's internal database. Using buttons such as "Next," "Previous," and "Select," the user can scroll through book names and confirm the required selection.

Once the book is selected, the system retrieves the corresponding rack number and shelf position from pre-programmed data.

2. Navigation Subsystem

The robot navigates through the library using line follower sensors. A predefined black line path is drawn on the floor connecting all racks. The infrared sensors detect the contrast between the black line and the surface, allowing the robot to move accurately toward the selected rack.

The navigation logic ensures:

- Forward movement toward racks
- Rack counting mechanism
- Controlled stopping at correct rack
- Smooth return to starting position

This structured path-based navigation makes the system simple, reliable, and suitable for controlled indoor environments.

3. Identification Subsystem (RFID-Based)

Each book in the library is equipped with an RFID tag containing a unique identification number (UID). When the robot reaches the designated rack, the RFID reader module scans the tags to identify the correct book.

The microcontroller compares the scanned UID with the stored UID of the selected book. Only when the correct match is found does the system proceed to activate the robotic arm for picking. This ensures high accuracy and prevents retrieval of incorrect books.

4. Robotic Arm Mechanism

The robotic arm is designed using multiple servo motors to provide precise control over movement. The arm performs the following actions:

- Base rotation toward the shelf

- Vertical lifting to reach different shelf heights
- Extension and positioning
- Gripping and holding the book securely

The servo motors allow angular control, making it possible to adjust the arm according to rack and shelf height. After gripping the book, the arm retracts safely before the robot moves back.

5. Motor Control Subsystem

The robot's movement is controlled using DC motors connected to an L298N motor driver module. The motor driver allows:

- Forward movement
- Backward movement
- Speed control
- Direction control

This ensures that the robot moves smoothly even while carrying the weight of the book.

6. Power Management System

A regulated power supply ensures stable voltage for all components. Since servo motors and DC motors require higher current than the microcontroller, proper voltage regulation is implemented to prevent fluctuations and system instability.

Overall Working Flow

The complete operation of the proposed system follows this sequence:

1. User selects the desired book from LCD interface.
2. Arduino retrieves rack and shelf information.
3. Robot starts navigation using line sensors.

4. Robot stops at the correct rack.
5. RFID reader scans and verifies the selected book.
6. Robotic arm adjusts to correct height.
7. Arm grips the book securely.
8. Robot returns to the starting point.

This systematic approach ensures reliability and repeatability in operation.

Advantages of the Proposed System

- Reduces manual searching time
- Minimizes human error
- Enhances accuracy in book retrieval
- Reduces librarian workload
- Demonstrates practical automation
- Cost-effective prototype model

2.3 Block Diagram of the System

The **Block Diagram of the Automated Book Picking Robot** represents the overall architecture of the system and shows how different hardware modules are interconnected and controlled by the central processing unit. The block diagram provides a simplified graphical representation of the system, helping to understand the flow of data and control signals between components.

At the center of the system is the **Arduino MEGA 2560 microcontroller**, which acts as the brain of the robot. All input devices, output devices, and control modules are connected to it.

Main Blocks of the System

The complete system consists of the following major blocks:

1. **Power Supply Unit**
2. **User Interface Unit**
3. **Microcontroller Unit (Arduino MEGA 2560)**
4. **Navigation System**
5. **Identification System (RFID Module)**
6. **Motor Driver Unit (L298N)**
7. **Robotic Arm (Servo Motors)**

1. Power Supply Unit

The power supply provides regulated voltage to all components.

- The Arduino board requires stable 5V.
- Servo motors and DC motors require higher current.
- A voltage regulator (buck converter) ensures safe and stable operation.

This block is essential to prevent voltage fluctuations and system instability.

2. User Interface Unit

This block includes:

- 16×2 LCD Display
- Push Buttons (Next, Previous, Select)

The user interface allows book selection. The selected book information is sent to the Arduino microcontroller for processing.

3. Microcontroller Unit – Arduino MEGA 2560

This is the central control unit. It performs the following functions:

- Reads input from buttons

- Displays data on LCD
- Controls line follower sensors
- Communicates with RFID module
- Sends control signals to motor driver
- Controls servo motors for robotic arm
- Executes navigation and picking algorithm

All system logic is programmed into this controller.

4. Navigation System (Line Follower Sensors)

The navigation block consists of infrared line sensors that detect the predefined path on the floor. These sensors send signals to the Arduino indicating whether the robot is on track or needs correction.

Based on sensor input:

- Arduino adjusts motor direction
- Robot moves toward the correct rack

5. RFID Identification System

This block includes:

- RFID Reader (RC522 Module)
- RFID Tags attached to books

When the robot reaches the selected rack, the RFID reader scans nearby tags. The Arduino compares the scanned UID with the stored UID of the selected book.

If matched:

- The robotic arm mechanism is activated.

6. Motor Driver Unit (L298N)

The L298N motor driver controls the DC motors used for robot movement.

It receives control signals from the Arduino and:

- Drives motors forward
- Drives motors backward
- Controls speed and direction

This block ensures proper movement of the robot base.

7. Robotic Arm (Servo Motors)

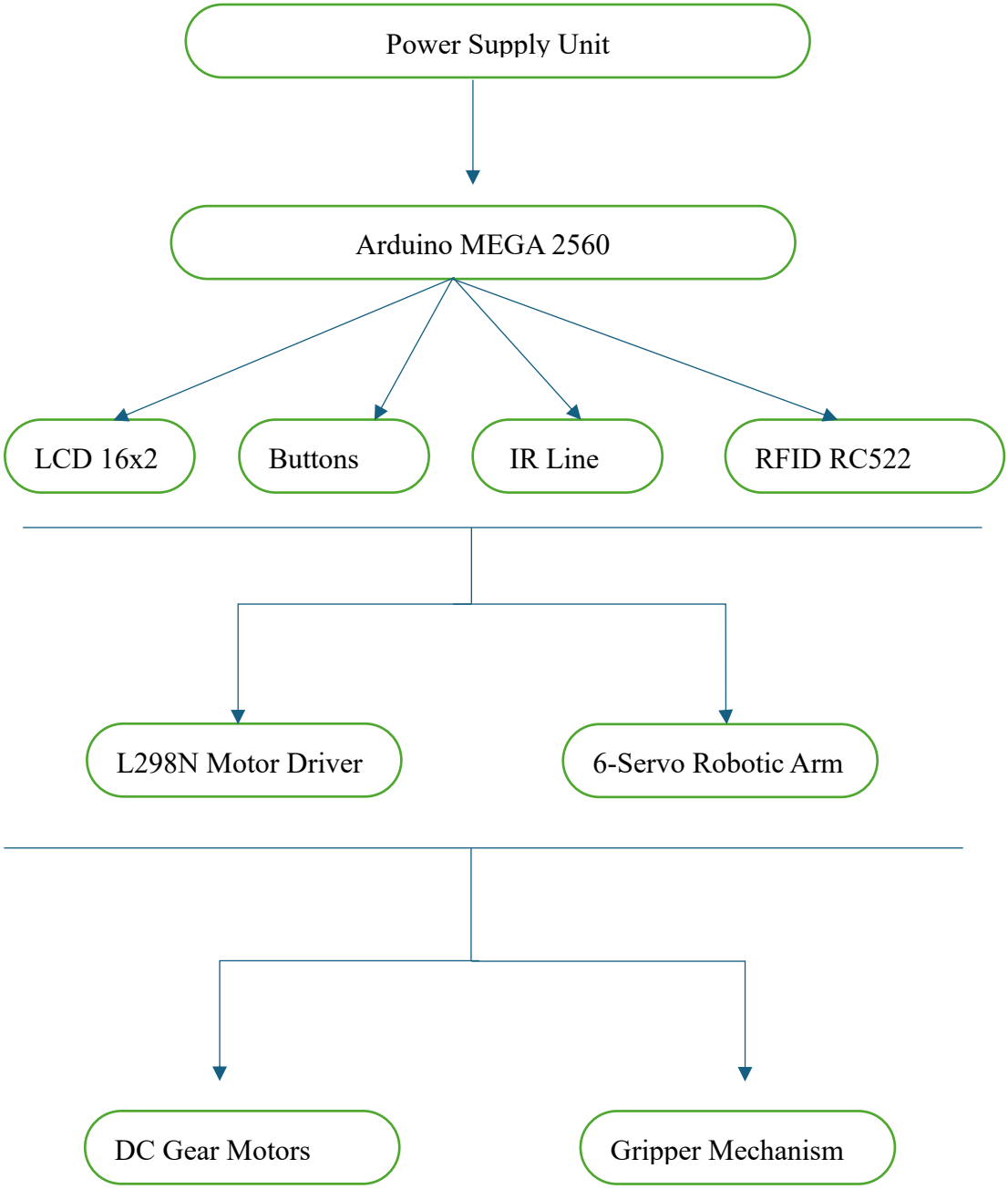
The robotic arm block consists of multiple servo motors responsible for:

- Base rotation
- Vertical lifting
- Arm extension
- Gripper control

The Arduino sends PWM signals to control the angular movement of each servo motor, allowing precise book picking.

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2.3 Block Diagram



2.4 Working Principle of the Robot

The **working principle of the Automated Book Picking Robot** is based on the integration of embedded control systems, sensor-based navigation, RFID identification, and servo-driven mechanical movement. The robot operates through a sequence of well-defined steps that ensure accurate book selection, navigation, identification, picking, and return.

The entire system is controlled by the **Arduino MEGA 2560 microcontroller**, which acts as the central processing unit. It continuously reads inputs from sensors and user controls, processes the data, and sends output signals to motors and actuators.

Step-by-Step Working Principle

Step 1: Book Selection

The process begins when the user interacts with the system through a **16×2 LCD display and push buttons**.

- The LCD displays the list of available books.
- The user scrolls using “Next” and “Previous” buttons.
- The desired book is selected using the “Select” button.

Once the selection is confirmed, the Arduino retrieves the stored information of that book, including:

- Rack number
- Shelf position
- RFID UID

Step 2: Navigation to the Rack

After selection, the robot begins moving toward the designated rack. The navigation system works using **line follower sensors** placed at the bottom of the robot.

- A black line path is drawn on the floor.

- Infrared sensors detect the black line.
- Based on sensor readings, the Arduino adjusts motor direction.
- The robot moves forward along the path.

The system may use rack counting logic (for example, detecting junctions or markers) to determine when the correct rack is reached.

Step 3: Rack Detection and Stopping

When the robot reaches the correct rack location:

- The Arduino stops the DC motors.
- The robot positions itself in front of the shelf.

Accurate stopping is achieved through programmed distance control or sensor-based detection.

Step 4: RFID Book Identification

Each book is equipped with an RFID tag containing a **Unique Identification Number (UID)**.

- The RFID reader module scans the tags on the shelf.
- The scanned UID is sent to the Arduino.
- The Arduino compares it with the selected book's stored UID.

If the UID matches:

- The system confirms that the correct book is identified.
- The robotic arm mechanism is activated.

If it does not match:

- The system continues scanning until the correct tag is found.

Step 5: Robotic Arm Activation

The robotic arm consists of multiple servo motors that provide precise angular control.

The arm performs the following actions:

1. Rotates toward the correct shelf.
2. Adjusts vertical height according to shelf position.
3. Extends forward toward the book.
4. Activates the gripper to hold the book securely.
5. Retracts safely.

Servo motors are controlled using PWM (Pulse Width Modulation) signals from the Arduino, ensuring smooth and accurate movements.

Step 6: Return to Starting Position

After successfully picking the book:

- The robot moves backward along the same path.
- It returns to its starting position.
- The book is delivered to the user.

This completes one full operation cycle.

Control Logic Principle

The working principle is based on a structured control sequence:

1. Input → User selection
2. Processing → Arduino algorithm execution
3. Navigation → Line sensor-based movement
4. Identification → RFID scanning and verification
5. Action → Servo motor-based picking
6. Output → Book retrieval and return

Each step is dependent on successful completion of the previous step, ensuring accuracy and reliability.

System Reliability

The robot operates in a controlled environment with:

- Predefined rack mapping
- Structured floor path
- Pre-programmed book database

This ensures repeatability and reduces chances of operational error.

CHAPTER 3 – SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The **Hardware Requirements** section describes all the physical components used in the development of the Automated Book Picking Robot. These components work together to enable navigation, identification, control, and book retrieval. Proper selection and integration of hardware components are essential to ensure system stability, efficiency, and reliability.

The hardware components used in this project can be categorized into the following groups:

1. Control Unit
2. Identification System
3. Navigation System
4. Motion Control System
5. Robotic Arm Mechanism
6. User Interface
7. Power Supply Unit
8. Mechanical Structure

1. Control Unit

Arduino MEGA 2560

The Arduino MEGA 2560 microcontroller acts as the central control unit of the robot. It processes all inputs and controls all outputs.

Key Features:

- 54 Digital I/O pins
- 16 Analog input pins
- 4 UART ports

- 16 MHz clock speed
- Large memory capacity

The Arduino Mega is chosen because this project requires multiple connections for servo motors, sensors, RFID module, LCD display, and motor driver simultaneously.

2. Identification System

RFID Module (RC522)

The RFID reader module is used to identify books through RFID tags.

Components:

- RC522 RFID Reader
- RFID Tags (attached to books)

Function:

- Reads Unique Identification Number (UID) from tags
- Sends UID data to Arduino
- Ensures accurate book verification

3. Navigation System

Line Follower Sensors

Infrared (IR) sensors are used to detect the black line path on the floor.

Function:

- Detects black/white surface difference
- Sends digital signals to Arduino
- Guides robot movement toward racks

Typically, 2 or 3 line sensors are used for accurate path tracking.

4. Motion Control System

DC Motors

DC motors are used to move the robot forward and backward.

L298N Motor Driver Module

The L298N motor driver controls the direction and speed of DC motors.

Functions:

- Controls motor direction (Forward/Reverse)
- Provides sufficient current to motors
- Isolates Arduino from high motor current

5. Robotic Arm Mechanism

Servo Motors

Multiple servo motors are used to control:

- Base rotation
- Arm lifting
- Extension
- Gripper opening and closing

Servo motors are selected because they provide accurate angular control using PWM signals.

Gripper Mechanism

The gripper holds the book securely without damaging it.

6. User Interface Components

16×2 LCD Display

Displays:

- Book names
- System status
- Operation messages

Push Buttons

Used for:

- Navigating book list
- Selecting desired book

7. Power Supply Unit

The system requires stable voltage for proper operation.

Battery Pack

Provides portable power supply.

Voltage Regulator / Buck Converter (CA2596)

Regulates voltage to:

- 5V for Arduino and sensors
- Suitable voltage for motors and servos

Proper grounding is essential to avoid system instability.

8. Mechanical Structure

Robot Chassis

- Base platform to mount components

- Supports motors and robotic arm

Wheels

- Two-wheel or four-wheel drive system
- Provides stable movement

Rack and Shelf Model

A structured rack model is built to simulate real library shelves.

3.2 Software Requirements

The **Software Requirements** section describes the programming tools, development platforms, and software components used to design, program, and control the Automated Book Picking Robot. Software plays a crucial role in coordinating all hardware components, implementing control logic, and ensuring smooth operation of the system.

The project is primarily based on **embedded system programming**, where the microcontroller executes instructions written in a high-level programming language. Proper software selection ensures system reliability, accurate communication between modules, and efficient performance.

1. Arduino IDE (Integrated Development Environment)

The main software used for programming the robot is the **Arduino IDE**.

Purpose:

- Writing embedded C/C++ code (Arduino sketches)
- Compiling and uploading code to Arduino MEGA 2560
- Serial monitoring for debugging
- Testing hardware communication

Features:

- Simple and user-friendly interface
- Built-in libraries for sensors and communication
- Cross-platform support (Windows, Linux, Mac)
- Real-time serial output monitoring

The Arduino IDE allows integration of external libraries such as:

- RFID library
- Servo motor library
- LCD display library

These libraries simplify programming and reduce development time.

2. Embedded C / C++ Programming

The control logic of the robot is written using **Embedded C/C++ language** within the Arduino environment.

Role of Programming:

- Reading sensor inputs
- Controlling motor direction and speed
- Generating PWM signals for servo motors
- Communicating with RFID module
- Displaying information on LCD
- Implementing navigation and picking algorithms

Embedded programming ensures real-time response to sensor inputs and precise control of robotic movements.

3. RFID Library

The RFID module requires a specialized software library for communication.

Functions of RFID Library:

- Initialize RFID reader
- Read tag UID
- Compare scanned UID with stored UID
- Handle data communication via SPI protocol

This library allows smooth integration between the RFID reader and Arduino

4. Servo Motor Library

Servo motors require PWM signal control for accurate angular positioning.

Functions of Servo Library:

- Attach servo to specific pin
- Control rotation angle (0°–180°)
- Provide smooth movement

The servo library simplifies robotic arm control and improves precision.

5. LCD Display Library

The 16×2 LCD display requires a display control library.

Functions:

- Initialize LCD
- Display text messages
- Clear and update screen
- Show book names and system status

This library improves user interaction and system communication.

6. Serial Monitor (Debugging Tool)

The Arduino IDE includes a **Serial Monitor** used for:

- Debugging code
- Monitoring sensor values
- Checking RFID UID data
- Verifying system performance

Debugging is essential to ensure correct system behavior.

7. Operating System Requirement

The development system requires:

- Windows 10 / Windows 11
- Minimum 4GB RAM
- USB port for Arduino connection

The operating system is used only for development and code uploading. The robot runs independently once programmed.

8. Algorithm Development

Software also includes the design of:

- Navigation algorithm
- Rack counting logic
- RFID verification logic
- Servo movement sequence
- Return-to-start routine

These algorithms ensure systematic and reliable operation of the robot.

3.3 Arduino MEGA 2560 – Description and Features

Description

Arduino MEGA 2560 is a powerful and versatile microcontroller board based on the **ATmega2560** microcontroller. It is designed for complex embedded applications that require a large number of input and output pins, increased memory, and reliable performance. In this project, the Arduino MEGA 2560 acts as the **main controller** of the Automatic Book Picking Robot, coordinating all sensing, decision-making, and actuation processes.

The Arduino MEGA 2560 receives input signals from push buttons, IR line sensors, and the RFID RC522 module. It processes this information based on the programmed logic and generates control signals for the motor driver, LCD display, and multiple servo motors of the robotic arm. Due to the involvement of many hardware components, the Arduino MEGA is chosen instead of Arduino UNO to avoid pin conflicts and to support multiple servos and peripherals simultaneouslyThe board operates at **5V logic level**, making it compatible with most commonly used sensors and modules. It supports serial communication protocols such as **UART, SPI, and I²C**, which are essential for interfacing with RFID modules, displays, and debugging tools.

Features of Arduino MEGA 2560

1. **Microcontroller:** ATmega2560
2. **Operating Voltage:** 5V
3. **Input Voltage (recommended):** 7–12V
4. **Digital I/O Pins:** 54 (15 pins support PWM)
5. **Analog Input Pins:** 16
6. **Flash Memory:** 256 KB (of which 8 KB used by bootloader)
7. **SRAM:** 8 KB
8. **EEPROM:** 4 KB
9. **Clock Speed:** 16 MHz
10. **Communication Interfaces:**

- 4 × UART (Serial ports)
- SPI
- I²C (TWI)

11. **USB Interface:** USB Type-B for programming and serial communication
12. **Power Jack:** External power support via DC jack
13. **On-board Voltage Regulator:** Provides stable power to the board
14. **Reset Button:** Allows manual system reset

Why Arduino MEGA 2560 is Used in This Project

- Supports **multiple servo motors** required for the robotic arm
- Provides sufficient I/O pins for **LCD, RFID, motors, sensors, and buttons**
- Handles **simultaneous operations** without pin conflicts
- Easy programming and wide community support
- Suitable for **large embedded robotics projects**

Role of Arduino MEGA 2560 in the Project

- Reads user input from buttons
- Displays system status on LCD
- Controls robot navigation using IR line sensors
- Communicates with RFID reader to identify books
- Drives DC motors via L298N motor driver
- Controls six servo motors of the robotic arm


MADE
IN ITALY
AREF
GND
PWM
COMMUNICATION
TX
RX
ON
RESET
ICSP
WWW.ARDUINO.CC
MEGA
ARDUINO
2560
POWER
ANALOG IN
DIGITAL

3.4 RFID TECHNOLOGY AND UID IDENTIFICATION

Introduction to RFID Technology

RFID (Radio Frequency Identification) is a wireless identification technology used to automatically identify objects using radio waves. An RFID system mainly consists of two components: an **RFID tag (card)** and an **RFID reader**. Unlike barcodes, RFID does not require direct line-of-sight contact and can identify objects quickly and accurately.

In this project, RFID technology is used to uniquely identify books in a library. Each book is attached with an RFID card, and an RFID reader mounted on the robotic arm scans the cards to identify the required book.

RFID RC522 Module

The **RC522 RFID module** is a commonly used low-cost RFID reader operating at **13.56 MHz** It supports ISO/IEC 14443 Type-A RFID cards. The module communicates with the Arduino MEGA using the **SPI communication protocol**.

Key features of RC522:

- Operates at 13.56 MHz
- Low power consumption
- Supports passive RFID cards
- Fast and reliable UID reading
- Compact size, suitable for robotic applications

RFID Tags / Cards

RFID tags used in this project are **passive RFID cards**, which do not require a battery. These cards receive power from the electromagnetic field generated by the RFID reader. Each card contains a small chip and antenna, which store a **Unique Identification Number (UID)**.

The cards used may look identical externally, but internally each card has a **different UID**, which makes it possible to distinguish one book from another even if they are placed on the same shelf.

UID (Unique Identification) Explanation

The **UID** is a unique number programmed into every RFID card during manufacturing. It cannot be changed or duplicated easily. The UID usually consists of **4 or 7 bytes**, represented in hexadecimal format.

Example UID:

A3 4F 9C 21

In this project:

- Each UID is stored in the Arduino program as part of the book database
- When the RFID reader scans a card, it reads the UID
- The Arduino compares the scanned UID with stored UIDs
- If a match is found, the corresponding book is identified

This method ensures **accurate identification** even when multiple books are present in the same rack and shelf.

RFID-Based Book Identification Process

1. RFID card is attached to each book
2. Robotic arm moves near the books on the shelf
3. RFID reader scans the cards one by one
4. UID is read and sent to Arduino MEGA
5. Arduino compares UID with stored database
6. Matching UID confirms the correct book

Advantages of Using RFID in This Project

- No physical contact required
- Fast and accurate identification
- Unique UID for every book
- Works even when cards are not visible
- Suitable for automated systems

Limitations

- Limited reading range (a few centimetres)
- Card orientation affects reading accuracy
- Requires RFID cards to be attached to books

Role of RFID Technology in the Project

RFID technology plays a critical role in ensuring that the robot selects and picks the **correct book**. It eliminates human error and allows the robot to differentiate between multiple books placed closely together.



3.5 LINE FOLLOWER SENSORS – WORKING AND LOGIC

Introduction

Line follower sensors are used to guide the robot along a predefined path. In the Automatic Book Picking Robot, **IR (Infrared) line follower sensors** are employed to navigate the robot accurately between library racks. These sensors help the robot move in a controlled manner without the need for complex navigation algorithms.

Construction of IR Line Follower Sensor

An IR line follower sensor consists of:

- **IR Transmitter (IR LED)** – emits infrared light
- **IR Receiver (Photodiode/Phototransistor)** – detects reflected IR light

The sensor works based on the **reflection principle** of infrared light from different surfaces.

Working Principle

- When IR light falls on a **white surface**, most of the light is reflected back and detected by the receiver.
- When IR light falls on a **black surface (line)**, most of the light is absorbed and very little light is reflected.

Based on the amount of reflected light, the sensor outputs a **digital signal**:

- **HIGH (1)** → White surface detected
- **LOW (0)** → Black line detected

In this project, a **black line on a white floor** is used as the navigation path.

Sensor Arrangement

Three-line follower sensors are used:

1. **Left Sensor**

2. **Centre Sensor**
3. **Right Sensor**

This arrangement allows the robot to:

- Move straight
- Correct left deviation
- Correct right deviation
- Detect rack markers

Logic Used for Line Following

The Arduino continuously reads the digital outputs of the three sensors and applies the following logic:

Left	Centre	Right	Robot Action
HIGH	LOW	HIGH	Move forward
LOW	HIGH	HIGH	Turn left
HIGH	HIGH	LOW	Turn right
LOW	LOW	LOW	Rack marker detected
HIGH	HIGH	HIGH	Stop / Line lost

Rack Identification Logic

Rack positions are marked using **horizontal black strips** on the line. When **all three sensors detect black (LOW)** simultaneously, it indicates that the robot has reached a rack marker.

- Each marker increments a **rack counter**
- When rack counter equals target rack number, the robot stops

This simple logic enables the robot to distinguish between **Rack 1, Rack 2, and Rack 3** accurately.

Advantages of Using Line Follower Sensors

- Simple and low-cost navigation method
- Reliable in structured environments
- Easy to implement and debug
- No need for complex processing

Limitations

- Works only on predefined paths
- Sensitive to lighting conditions
- Cannot avoid dynamic obstacles

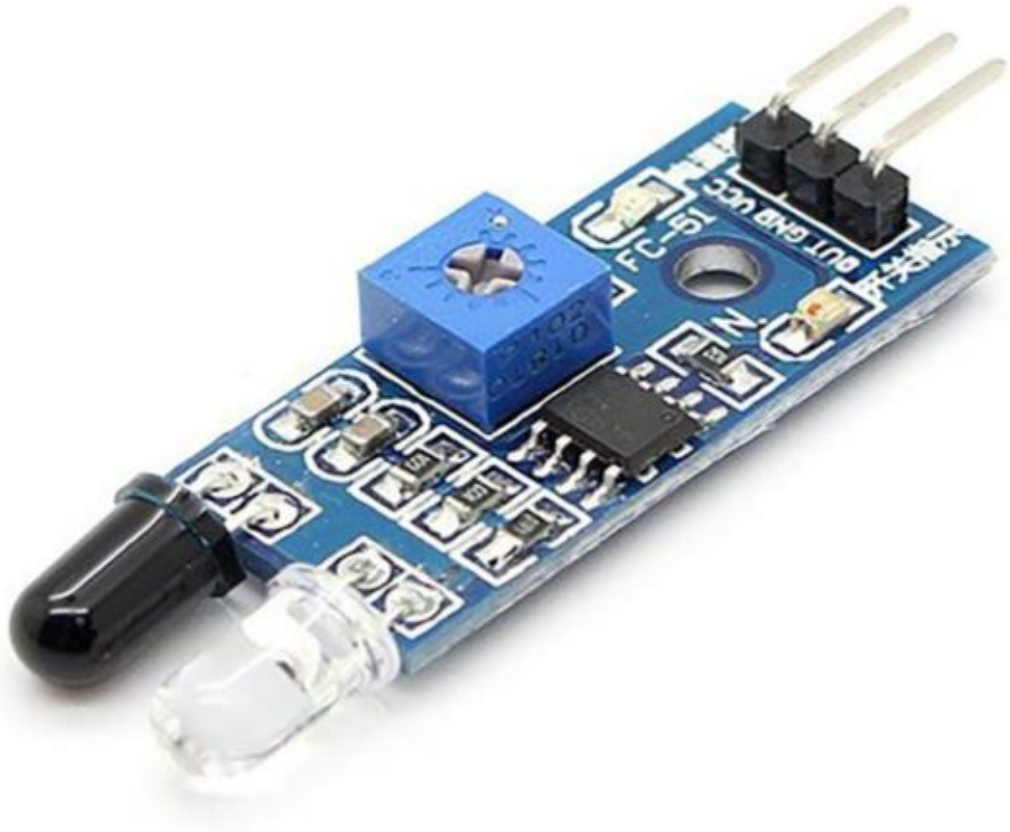
Role of Line Follower Sensors in the Project

Line follower sensors are responsible for:

- Guiding the robot between racks
- Ensuring accurate navigation
- Identifying rack positions using markers

Without these sensors, the robot would not be able to reach the correct rack automatically.

IR Sensor Complete Guide



3.6 MOTOR DRIVER (L298N) AND MOTOR CONTROL

Introduction

In robotic systems, motors require higher current and voltage than what a microcontroller can supply directly. Therefore, a motor driver is essential to interface the controller with the motors. In this project, the **L298N motor driver module** is used to control the DC gear motors that move the robot chassis forward and backward.

L298N Motor Driver Overview

The **L298N** is a **dual H-bridge motor driver** capable of controlling two DC motors independently. It allows control of both **direction** and **speed** of motors using digital and PWM signals from the Arduino MEGA 2560.

Key features of L298N:

- Dual H-bridge motor driver
- Can control two DC motors simultaneously
- Supports motor voltage up to 46V
- Provides current up to 2A per channel
- Direction and speed control using logic inputs

Why L298N is Used in This Project

- Arduino cannot directly drive motors due to low current output
- L298N provides sufficient current and voltage to motors
- Simple interface and widely available
- Suitable for educational and prototype robotic applications

Motor Connections

The robot uses **four DC gear motors** arranged in a four-wheel chassis:

- **Left Front and Left Rear motors** are connected in parallel to **OUT1 & OUT2**
- **Right Front and Right Rear motors** are connected in parallel to **OUT3 & OUT4**

This configuration allows synchronized movement of motors on each side.

Control Pins and Their Function

- **IN1 & IN2** → Control direction of left motors
- **IN3 & IN4** → Control direction of right motors
- **ENA** → Enable and speed control for left motors (PWM)
- **ENB** → Enable and speed control for right motors (PWM)

PWM signals from the Arduino are applied to ENA and ENB pins to control motor speed.

Motor Direction Control Logic

IN1	IN2	IN3	IN4	Robot Action
HIGH	LOW	HIGH	LOW	Move Forward
LOW	HIGH	LOW	HIGH	Move Backward
LOW	LOW	LOW	LOW	Stop
LOW	HIGH	HIGH	LOW	Turn Left
HIGH	LOW	LOW	HIGH	Turn Right

Working Principle of Motor Control

The Arduino MEGA sends control signals to the L298N motor driver based on input from line follower sensors and system logic. When the robot needs to move forward, the Arduino sets the appropriate input pins HIGH and LOW, causing the motors to rotate in the forward direction. For backward motion, the pin logic is reversed. Speed is controlled using PWM signals on the enable pins

Power Supply Considerations

The L298N motor driver is powered directly from the battery supply to meet motor voltage requirements. The ground of the L298N is connected to the Arduino ground to maintain a common reference. Proper power distribution ensures smooth motor operation without affecting the controller.

Advantages of Using L298N

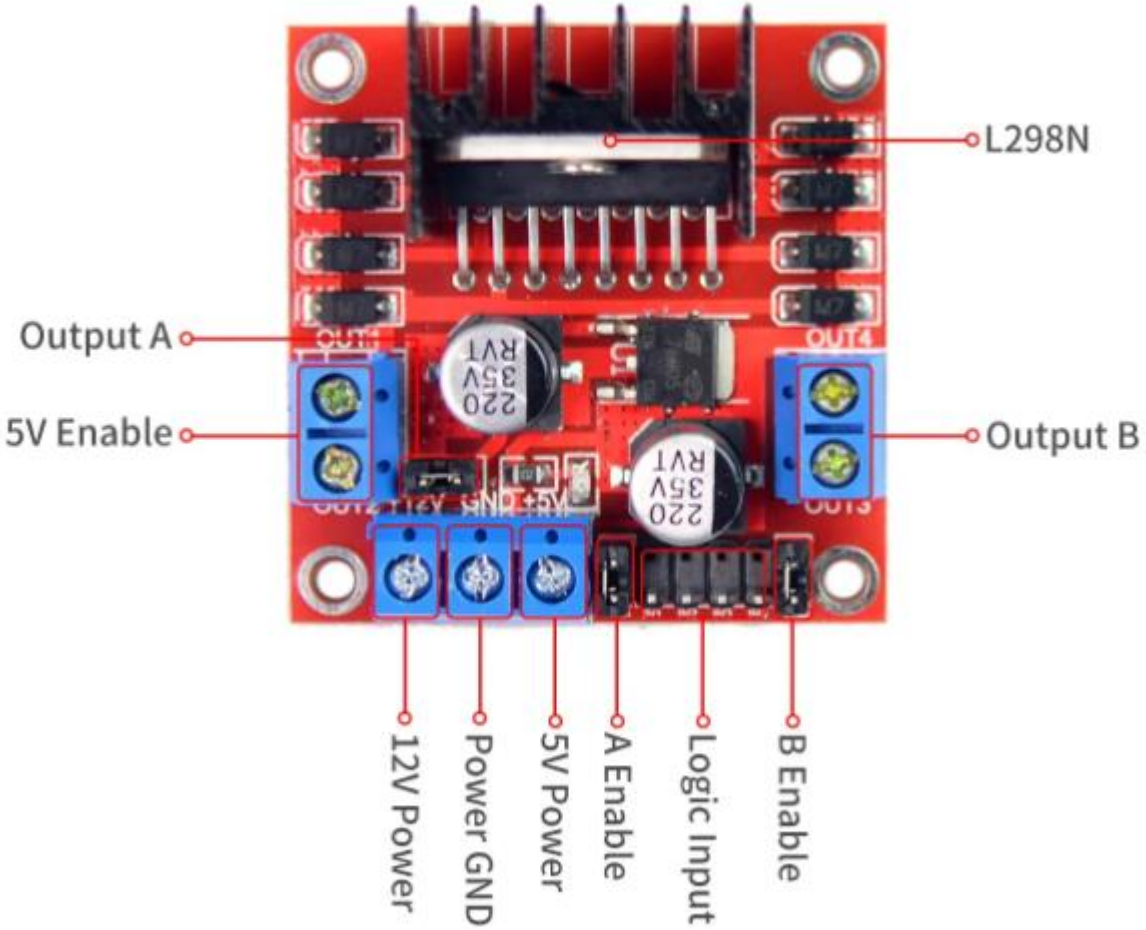
- Simple motor control logic
- Reliable and robust operation
- Supports bidirectional motor control
- Easy integration with Arduino

Limitations

- Lower efficiency due to internal voltage drop
- Generates heat at higher loads
- Not suitable for very high-current motors

Role of L298N in the Project

The L298N motor driver plays a crucial role in enabling controlled navigation of the robot. It allows the robot to move accurately between racks, stop at precise locations, and return safely to the starting point.


L298N
Output A
5V Enable
Output B
12V Power
Power GND
5V Power
A Enable
Logic Input
B Enable

CHAPTER 4 – SYSTEM DESIGN

4.1 ROBOTIC ARM DESIGN USING SERVO MOTORS

Introduction

The robotic arm is one of the most important parts of the Automatic Book Picking Robot. Its main function is to reach the correct shelf, identify the required book, and pick it safely. In this project, a **servo-motor-based robotic arm** is designed to provide accurate and controlled movement in a limited space, which is ideal for library environments.

Design Concept

The robotic arm is designed as a **multi-degree-of-freedom (DOF) mechanism** driven by **six servo motors**. Each servo motor controls one specific joint or movement of the arm. Servo motors are selected because they provide precise angular control, require simple control signals, and are easy to interface with Arduino.

The arm is mounted on the robot chassis, and the RFID reader is attached near the gripper to scan book RFID cards effectively.

Servo Motors Used and Their Functions

The six servo motors used in the robotic arm perform the following functions:

1. **Base Rotation Servo**
   Controls left-to-right rotation of the arm. It allows the arm to scan multiple books on the same shelf.
2. **Shoulder Servo**
   Controls vertical movement of the arm. It is used to adjust the arm height according to the shelf level.
3. **Elbow Servo**
   Controls forward and backward extension of the arm. It helps the arm reach books placed deeper on the shelf.

4. **Wrist Rotation Servo**
   Rotates the wrist to align the gripper properly with the book.
5. **Grip Position (Tilt) Servo**
   Adjusts the angle of the gripper to ensure proper orientation before picking the book.
6. **Gripper Open/Close Servo**
   Controls the opening and closing of the gripper jaws to hold and release the book securely.

Working of the Robotic Arm

After the robot reaches the correct rack, the shoulder servo moves the arm to the required shelf height. The base rotation servo then sweeps the arm horizontally across the shelf to scan RFID tags. When the RFID reader detects a matching UID, the elbow servo moves the arm forward, the grip position servo aligns the gripper, and the gripper servo closes to pick the book. The arm then retracts to a safe position.

Servo Control Logic

Each servo motor is controlled using PWM signals generated by the Arduino MEGA. The servo angle is set by sending values between **0° and 180°**, allowing smooth and precise motion. The use of multiple servos enables coordinated and accurate movement of the robotic arm.

Power Considerations

Servo motors require higher current than the Arduino can supply. Therefore, all servos are powered from an external regulated power source using a buck converter, while sharing a common ground with the Arduino. This ensures stable and reliable arm operation.

Advantages of Servo-Based Robotic Arm

- Precise angular control
- Simple interfacing with Arduino
- Compact and lightweight design
- Suitable for pick-and-place operations
- Low cost and easy maintenance

Role of the Robotic Arm in the Project

The robotic arm enables the robot to physically interact with the environment by reaching shelves, scanning books, and picking the correct book accurately. Without the robotic arm, the robot would not be able to retrieve books automatically.


SG90 Micro Servo
MG996R Servo

4.2 POWER SUPPLY AND VOLTAGE REGULATION (CA2596)

Introduction

A reliable and stable power supply is essential for the proper functioning of any robotic system. In the Automatic Book Picking Robot, multiple components such as the Arduino MEGA 2560, DC motors, servo motors, RFID reader, LCD, and sensors require different voltage and current levels. To meet these requirements, a regulated power supply system using **lithium-ion batteries** and a **CA2596 buck converter** is employed.

Battery Power Source

The robot is powered using **two 3.7V lithium-ion batteries connected in series**, providing a total voltage of approximately **7.4V to 8.4V** when fully charged. Lithium-ion batteries are chosen because they are lightweight, rechargeable, and capable of delivering sufficient current for motors and servo motors.

Need for Voltage Regulation

Different components in the robot operate at different voltage levels:

- Arduino MEGA and sensors → 5V
- Servo motors → 5V to 6V (high current)
- DC motors → Battery voltage (7–12V)
- RFID RC522 → 3.3V

Directly supplying battery voltage to these components may damage them. Therefore, voltage regulation is required.

CA2596 Buck Converter

The **CA2596** is a DC-DC buck (step-down) voltage converter used to reduce the battery voltage to a stable and safe level. It provides high efficiency and can handle higher current compared to linear regulators.

Key features of CA2596:

- Adjustable output voltage
- High efficiency (up to 90%)
- Can supply high current for servo motors
- Compact and reliable

Working Principle of CA2596

The CA2596 takes the higher input voltage from the battery and steps it down to a lower regulated voltage using switching regulation. The output voltage is adjusted using an onboard potentiometer and verified with a multimeter before connecting to the circuit.

In this project:

- CA2596 output is set to **5.5V**
- This regulated voltage is used to power the Arduino MEGA and servo motors

Power Distribution in the Robot

- **Battery directly powers the L298N motor driver** for DC motors
- **CA2596 supplies regulated voltage** to Arduino MEGA, servo motors, LCD, and sensors
- **Arduino 3.3V pin** is used to power the RFID RC522 module
- **All grounds (GND) are connected together** to maintain a common reference

Advantages of Using CA2596

- Protects components from overvoltage
- Provides stable voltage under load
- Improves system reliability
- Efficient and reduces power loss as heat

Safety Considerations

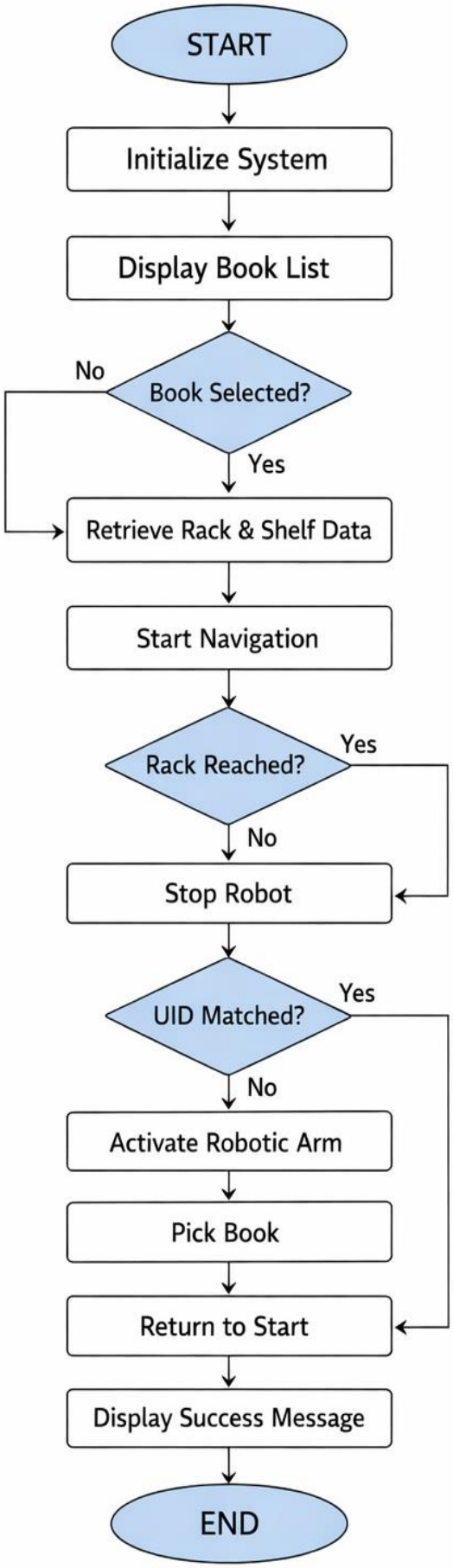
- Output voltage must be adjusted before connecting components
- Use proper wiring to handle servo motor current
- Ensure common ground between all modules
- Avoid powering servo motors directly from Arduino 5V pin

Role of Power Supply in the Project

The power supply system ensures smooth operation of the robot by providing the required voltage and current to each component. Stable power regulation is critical for accurate motor control, reliable RFID reading, and safe robotic arm operation.



4.3 Flowchart of System Operation



Flowchart Explanation

1. **Start & Initialization**

   When power is supplied, the Arduino MEGA initializes all hardware modules such as LCD, buttons, motors, servo motors, RFID reader, and line follower sensors.

2. **Book Selection**

   The user selects the required book using push buttons. The selected book name is displayed on the LCD.

3. **Book Confirmation**

   Once the user confirms the selection, the system retrieves the corresponding rack number, shelf number, and RFID UID from the database.

4. **Navigation to Rack**

   The robot moves forward by following a black line using IR line sensors. Rack markers are detected and counted to identify the target rack.

5. **Rack Reached**

   When the target rack count matches the stored rack number, the robot stops.

6. **Shelf Positioning**

   The robotic arm adjusts its height using servo motors to reach the required shelf.

7. **RFID Scanning**

   The arm rotates left and right while the RFID reader scans the RFID cards attached to the books.

8. **Book Identification**

   The scanned UID is compared with the stored UID. If there is no match, scanning continues.

9. **Book Picking**

   When a match is found, the robotic arm grips the book securely using the gripper.

10. **Return to Start Position**

    The robot moves backward along the same path to reach the starting position.

11. **Book Delivery**

    The book is released for the user to collect, and the LCD displays a confirmation message.

12. **System Reset**

    The system resets and waits for the next book selection.

4.4 ALGORITHM AND CONTROL LOGIC

Introduction

The algorithm and control logic define the sequence of operations performed by the Automatic Book Picking Robot. The control logic ensures proper coordination between user input, navigation, book identification, robotic arm movement, and system reset. The entire operation is controlled by the Arduino MEGA 2560 using a predefined algorithm.

STEP-BY-STEP ALGORITHM

Step 1: System Initialization

1. Start the system.
2. Initialize Arduino MEGA, LCD display, push buttons, IR line sensors, RFID reader, motor driver, and servo motors.
3. Display “Select Book” on the LCD.

Step 2: Book Selection

4. Read user input from NEXT and PREVIOUS buttons.
5. Display selected book name on the LCD.
6. If SELECT button is pressed, confirm the book selection.
7. Retrieve corresponding rack number, shelf number, and RFID UID from the database.

Step 3: Navigation to Rack

8. Activate DC motors through L298N motor driver.
9. Follow the black line using IR line follower sensors.
10. Continuously monitor sensor outputs.
11. Detect rack markers when all sensors read LOW.

12. Increment rack counter.
13. Stop the robot when rack counter equals target rack number.

Step 4: Shelf Position Adjustment

14. Move robotic arm to the required shelf height using shoulder servo motor.
15. Stabilize the robot at the rack.

Step 5: Book Scanning and Identification

16. Rotate the robotic arm left and right using base and wrist servos.
17. Activate RFID reader to scan RFID cards on books.
18. Read UID from RFID card.
19. Compare scanned UID with stored UID.
20. If UID does not match, continue scanning.
21. If UID matches, confirm correct book identification.

Step 6: Book Picking Operation

22. Move elbow servo to reach the book.
23. Adjust grip position using grip tilt servo.
24. Close gripper servo to hold the book securely.
25. Retract robotic arm to safe position.

Step 7: Return to Starting Position

26. Move robot backward along the same path.
27. Stop the robot at the starting position.

28. Release the book for the user.

Step 8: System Reset

29. Reset rack counter and flags.

30. Display “Select Book” on the LCD.

31. Wait for next user input.

4.5 Arduino Program

Coding

```
/************************************************************
 * AUTOMATIC BOOK PICKING ROBOT – FINAL VERSION
 * Board  : Arduino MEGA 2560
 * Design : Small space demo
 * Motion : Forward & backward only
 * Search : Rotating arm + RFID
 * Shelf  : Vertical lift servo control
 * Arm    : 6 Servo Motors
 ************************************************************/

#include <SPI.h>
#include <MFRC522.h>
#include <LiquidCrystal.h>
#include <Servo.h>

/* ================= FUNCTION DECLARATIONS ================= */
void selectBook();
void moveToRack();
void followLineForward();
void moveBackwardToStart();
void moveToShelf(int shelf);
void scanBooksWithArm();
```

```
void pickBook();


void moveForward();
void moveBackward();
void stopMotors();


/* ================== RFID RC522 PINS (MEGA SPI) ================== */
#define SS_PIN   53
#define RST_PIN  49
MFRC522 rfid(SS_PIN, RST_PIN);


/* ================== LCD PINS (MEGA) ================== */
// RS, E, D4, D5, D6, D7
LiquidCrystal lcd(30, 31, 32, 33, 34, 35);


/* ================== BUTTON PINS ================== */
#define BTN_NEXT    40
#define BTN_PREV    41
#define BTN_SELECT  42


/* ================== LINE SENSOR PINS ================== */
#define LINE_LEFT    A0
#define LINE_CENTER  A1
#define LINE_RIGHT   A2
```

```
/* ================== L298N MOTOR DRIVER ================== */

#define IN1 22

#define IN2 23

#define IN3 24

#define IN4 25

#define ENA 6

#define ENB 7


/* ================== 6 SERVO MOTORS ================== */

Servo baseServo;       // base rotation

Servo shoulderServo;   // shoulder up/down (shelf height)

Servo elbowServo;      // elbow movement

Servo wristRotate;     // wrist rotation

Servo gripTilt;        // grip position (tilt)

Servo gripperServo;    // open / close gripper


#define BASE_SERVO_PIN      8

#define SHOULDER_SERVO_PIN  9

#define ELBOW_SERVO_PIN     10

#define WRIST_SERVO_PIN     11

#define GRIP_TILT_PIN       12

#define GRIPPER_SERVO_PIN  13
```

```
/* ================== BOOK DATABASE ================== */

struct Book {
  byte uid[4];
  int rack;
  int shelf;
  const char* name;
};

Book books[] = {
  {{0xE, 0x4B, 0xE2, 0x0}, 1, 1, "AI Basics"},
  {{0xB2, 0x1A, 0x33, 0x90}, 2, 2, "Robotics"},
  {{0xC1, 0x55, 0x88, 0x44}, 3, 3, "IoT"}
};

int totalBooks = sizeof(books) / sizeof(books[0]);
int selectedBook = 0;
int targetRack = 0;
int rackCount = 0;
bool bookConfirmed = false;

/* ================== SETUP ================== */

void setup() {
  Serial.begin(9600);
```

```
lcd.begin(16, 2);
lcd.print("Smart Library");
delay(2000);
lcd.clear();

pinMode(BTN_NEXT, INPUT_PULLUP);
pinMode(BTN_PREV, INPUT_PULLUP);
pinMode(BTN_SELECT, INPUT_PULLUP);

pinMode(LINE_LEFT, INPUT);
pinMode(LINE_CENTER, INPUT);
pinMode(LINE_RIGHT, INPUT);

pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);
pinMode(IN3, OUTPUT);
pinMode(IN4, OUTPUT);
pinMode(ENA, OUTPUT);
pinMode(ENB, OUTPUT);

baseServo.attach(BASE_SERVO_PIN);
shoulderServo.attach(SHOULDER_SERVO_PIN);
elbowServo.attach(ELBOW_SERVO_PIN);
wristRotate.attach(WRIST_SERVO_PIN);
```

```
  gripTilt.attach(GRIP_TILT_PIN);
  gripperServo.attach(GRIPPER_SERVO_PIN);

  baseServo.write(90);
  shoulderServo.write(50);
  elbowServo.write(90);
  wristRotate.write(90);
  gripTilt.write(90);
  gripperServo.write(90);

  SPI.begin();
  rfid.PCD_Init();

  lcd.print("Select Book");
}

/* ================== MAIN LOOP ================== */
void loop() {
  if (!bookConfirmed) {
    selectBook();
  } else {
    moveToRack();
    moveToShelf(books[selectedBook].shelf);
    scanBooksWithArm();
```

```
    moveBackwardToStart();
    bookConfirmed = false;
    rackCount = 0;
    lcd.clear();
    lcd.print("Select Book");
  }
}

/* ================= BOOK SELECTION ================= */
void selectBook() {
  lcd.setCursor(0, 0);
  lcd.print("Select Book     ");
  lcd.setCursor(0, 1);
  lcd.print(books[selectedBook].name);

  if (digitalRead(BTN_NEXT) == LOW) {
    selectedBook = (selectedBook + 1) % totalBooks;
    delay(300);
  }
  if (digitalRead(BTN_PREV) == LOW) {
    selectedBook = (selectedBook - 1 + totalBooks) % totalBooks;
    delay(300);
  }
  if (digitalRead(BTN_SELECT) == LOW) {
```

```
    targetRack = books[selectedBook].rack;
    bookConfirmed = true;
    lcd.clear();
    lcd.print("Book Selected");
    delay(1000);
  }
}

/* ================= MOVE TO RACK ================= */
void moveToRack() {
  lcd.clear();
  lcd.print("Moving to Rack");

  while (rackCount < targetRack) {
    followLineForward();
    if (digitalRead(LINE_LEFT) == LOW &&
        digitalRead(LINE_CENTER) == LOW &&
        digitalRead(LINE_RIGHT) == LOW) {
      rackCount++;
      delay(600);
    }
  }
  stopMotors();
}
```

```
/* ================== LINE FOLLOW FORWARD ================== */
void followLineForward() {
  int L = digitalRead(LINE_LEFT);
  int C = digitalRead(LINE_CENTER);
  int R = digitalRead(LINE_RIGHT);

  if (C == LOW) moveForward();
  else if (L == LOW) {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);
  }
  else if (R == LOW) {
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
  }
  else stopMotors();
}

/* ================== MOVE TO SHELF HEIGHT ================== */
void moveToShelf(int shelf) {
  lcd.clear();
  lcd.print("Shelf ");
  lcd.print(shelf);
```

```
  if (shelf == 1) shoulderServo.write(50);
  if (shelf == 2) shoulderServo.write(75);
  if (shelf == 3) shoulderServo.write(100);

  delay(800);
}

/* ================= ARM SCAN & RFID ================= */
void scanBooksWithArm() {
  lcd.clear();
  lcd.print("Scanning Books");

  for (int angle = 40; angle <= 140; angle += 20) {
    baseServo.write(angle);
    wristRotate.write(angle);
    delay(600);

    if (!rfid.PICC_IsNewCardPresent()) continue;
    if (!rfid.PICC_ReadCardSerial()) continue;

    bool match = true;
    for (byte i = 0; i < 4; i++) {
      if (rfid.uid.uidByte[i] != books[selectedBook].uid[i]) {
```

```
        match = false;
        break;
      }
    }

    if (match) {
      lcd.clear();
      lcd.print("Book Found");
      pickBook();
      break;
    }
  }
}

/* ================== PICK BOOK ================== */
void pickBook() {
  elbowServo.write(60);
  delay(800);
  gripTilt.write(100);
  delay(400);
  gripperServo.write(10);
  delay(800);
  elbowServo.write(90);
}
```

```
/* ================== RETURN TO START ================== */
void moveBackwardToStart() {
  lcd.clear();
  lcd.print("Returning");

  moveBackward();
  delay(3000);
  stopMotors();

  lcd.clear();
  lcd.print("Collect Book");
}

/* ================== MOTOR CONTROL ================== */
void moveForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
  analogWrite(ENA, 150);
  analogWrite(ENB, 150);
}

void moveBackward() {
```

```
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
  analogWrite(ENA, 150);
  analogWrite(ENB, 150);
}

void stopMotors() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
```

CHAPTER 5

5.1 TESTING AND CALIBRATION OF COMPONENTS

Introduction

Testing and calibration are essential steps to ensure the reliable and accurate operation of the Automatic Book Picking Robot. Each hardware component is tested individually before system integration, and calibration is performed to fine-tune sensor readings, motor speeds, and servo positions. Proper testing helps identify faults early and ensures smooth overall system performance.

1. Power Supply Testing and Calibration

The power supply is tested first to ensure safe voltage levels for all components.

- Battery voltage is measured using a multimixer.
- The CA2596 buck converter output is adjusted using its onboard potentiometer.
- Output voltage is set to **5–6V** for Arduino, sensors, and servo motors.
- Common ground connection is verified across all modules.

This step prevents overvoltage damage and ensures stable operation under load.

2. Arduino MEGA 2560 Testing

The Arduino board is tested by:

- Uploading a basic LED blink program to verify board functionality.
- Checking serial communication using the Serial Monitor.
- Verifying correct pin operation by toggling digital and analogy pins.

This confirms that the controller is functioning correctly before connecting peripherals.

3. IR Line Follower Sensor Calibration

Line follower sensors are calibrated to accurately detect the black line.

- Sensors are placed over black and white surfaces.
- Sensitivity is adjusted using the onboard potentiometer (if available).
- Output logic is verified:
    - **LOW** → Black line detected
    - **HIGH** → White surface detected

Calibration ensures reliable navigation and accurate rack detection.

4. DC Motor and L298N Driver Testing

Motor testing is performed using a simple motor control program.

- Forward, backward, and stop motions are tested.
- Motor direction is verified for correct wiring.
- PWM values are adjusted to control speed.
- Left and right motors are synchronized to ensure straight movement.

This step ensures smooth and controlled robot motion.

5. Servo Motor Testing and Calibration

Each servo motor is tested individually:

- Servo test code is used to rotate servos from 0° to 180°.
- Neutral positions (90°) are identified.
- Mechanical alignment is adjusted to avoid over-rotation.
- Final angle values are calibrated for:
    - Shelf heights

- Arm extension
- Gripper open/close positions

Calibration prevents mechanical strain and improves accuracy.

6. RFID RC522 Module Testing

RFID testing is done using a UID reading program.

- RFID cards are scanned to read UID values.
- UID values are displayed on the Serial Monitor.
- Multiple cards are tested to confirm unique identification.
- Reading distance and orientation are checked.

This ensures accurate and reliable book identification.

7. LCD Display Testing

The LCD is tested by:

- Displaying static text and scrolling messages.
- Adjusting the contrast using a potentiometer.
- Verifying proper character visibility.

This confirms correct wiring and display functionality.

8. Button Switch Testing

Button testing includes:

- Verifying INPUT_PULLUP configuration.
- Checking response for NEXT, PREVIOUS, and SELECT buttons.
- Ensuring no false triggering or bouncing.

This ensures smooth user interaction.

9. Integrated System Testing

After individual testing, all components are integrated.

- Robot navigation is tested on a predefined line path.
- Rack detection logic is verified.
- Robotic arm scanning and picking operations are tested.
- Return-to-start motion is validated.

System-level testing ensures coordinated operation.

..

5.2 EXPERIMENTAL RESULTS AND OBSERVATIONS

Introduction

After successful design, implementation, and testing, the Automatic Book Picking Robot was experimentally evaluated in a controlled library-like environment. The experiments were conducted to verify navigation accuracy, book identification reliability, robotic arm performance, and overall system coordination.

1. Navigation Performance

Result:

The robot successfully followed the predefined black line using IR line follower sensors and moved smoothly between racks.

Observation:

- The robot maintained stable forward and backward motion.
- Rack markers were correctly detected when all three sensors sensed the black strip.
- Rack counting logic accurately identified Rack 1, Rack 2, and Rack 3.

Inference:

Line-following navigation proved to be reliable and suitable for structured environments like libraries.

2. Book Selection and User Interface

Result:

The user was able to select books using push buttons, and the selected book name was displayed correctly on the LCD.

Observation:

- Button response was quick and accurate.
- LCD displayed book names and system status messages clearly.
- No incorrect selection was observed during testing.

Inference:

The user interface is simple, effective, and suitable for non-technical users.

3. Shelf Positioning Accuracy

Result:

The robotic arm adjusted correctly to different shelf heights using servo motors.

Observation:

- Shelf 1, Shelf 2, and Shelf 3 positions were reached consistently.
- Minor calibration improved positioning accuracy.

Inference:

Servo-based shelf positioning is effective for small-scale library setups.

4. RFID-Based Book Identification

Result:

The RFID RC522 reader successfully detected and identified the correct book using UID matching.

Observation:

- RFID cards were read accurately when placed within the effective reading range.
- UID comparison correctly distinguished one book from multiple books on the same shelf.
- Orientation of the RFID card affected reading distance.

Inference:

RFID provides reliable and accurate book identification when properly positioned.

5. Robotic Arm and Gripper Performance

Result:

The robotic arm successfully picked and held the selected book.

Observation:

- All six servo motors operated smoothly.
- Gripper securely held the book without slippage.
- Arm returned safely to the default position after picking.

Inference:

The multi-servo robotic arm design is effective for pick-and-place operations.

6. Return-to-Start Operation

Result:

The robot returned to the starting position after book retrieval.

Observation:

- Backward motion was consistent and controlled.
- Robot stopped correctly at the starting point.

Inference:

Backward navigation is a simple and reliable method for limited-space environments.

7. Power Supply Stability

Result:

The power supply system remained stable during operation.

Observation:

- CA2596 provided regulated voltage under load.
- No unexpected resets or voltage drops were observed.

Inference:

Proper voltage regulation is critical for reliable robotic operation.

Overall System Performance

The robot successfully completed the full operation cycle:

- Book selection
- Navigation to rack
- Shelf positioning
- RFID scanning
- Book picking
- Return and delivery

The system operated reliably within the defined constraints

5.3 ADVANTAGES OF THE PROPOSED SYSTEM

The Automatic Book Picking Robot offers several advantages over traditional manual book retrieval systems. By integrating embedded systems, sensors, RFID technology, and robotics, the proposed system improves efficiency, accuracy, and usability in library environments.

1. Reduced Human Effort

The system minimizes manual involvement in searching and retrieving books. Library staff and users no longer need to physically locate books across multiple racks and shelves, reducing workload and fatigue.

2. Time Efficiency

Automated navigation and RFID-based identification significantly reduce the time required to retrieve a book. The robot can quickly locate the correct rack, shelf, and book without unnecessary searching.

3. Accurate Book Identification

Using RFID technology ensures precise identification of books. Each book has a unique UID, which eliminates confusion between books with similar titles or appearance.

4. User-Friendly Operation

The system provides a simple interface using push buttons and an LCD display. Users can easily select books and receive real-time status updates without technical knowledge.

5. Low-Cost and Affordable Design

The use of Arduino MEGA, low-cost sensors, and readily available components makes the system affordable and suitable for educational institutions and small libraries.

6. Reduced Human Errors

Automation eliminates common errors such as misreading labels, incorrect shelving, and selecting the wrong book.

7. Consistent and Reliable Performance

Unlike humans, the robot operates without fatigue and provides consistent performance throughout repeated operations.

8. Scalability

The system can be easily expanded to support more racks, shelves, and books by updating the database and navigation markers.

9. Safe Book Handling

The robotic arm and gripper are designed to handle books gently, reducing the risk of damage during retrieval.

10. Educational Value

The project demonstrates practical applications of robotics, embedded systems, and automation, making it ideal for academic learning and demonstrations.

11. Energy Efficient Operation

The use of rechargeable batteries and a buck converter ensures efficient power usage and portability.

12. Foundation for Smart Libraries

The proposed system serves as a base for future smart library solutions and can be enhanced with advanced technologies.

5.4 LIMITATIONS AND FUTURE ENHANCEMENTS

Limitations of the Proposed System

Although the Automatic Book Picking Robot performs effectively as a prototype system, it has certain limitations that should be considered:

1. **Predefined Path Requirement**
   The robot relies on a predefined black line for navigation. Any change in library layout requires re-laying the line path.
2. **Limited Operating Environment**
   The system is suitable only for controlled and structured environments. It cannot handle dynamic obstacles such as people moving in the path.
3. **RFID Reading Range Limitation**
   The RC522 RFID reader has a short reading range. Proper placement and orientation of RFID cards are required for reliable detection.
4. **Manual Book Database Update**
   Book details such as UID, rack number, and shelf number must be manually updated in the Arduino program.
5. **Mechanical Constraints**
   The robotic arm is designed for lightweight books only. Heavy or oversized books may not be handled safely.
6. **No Obstacle Avoidance**
   The current system does not include obstacle detection sensors such as ultrasonic or IR proximity sensors.
7. **Battery Dependency**
   The operating time of the robot is limited by battery capacity and requires periodic recharging.
8. **Single-Robot Operation**
   The system supports only one robot and does not include coordination between multiple robots.

Future Enhancements

The proposed system can be further improved by implementing the following enhancements:

1. **Obstacle Avoidance System**
   Integrating ultrasonic or LiDAR sensors to detect and avoid obstacles dynamically.
2. **Wireless Communication**
   Adding Wi-Fi or Bluetooth to allow book selection through a mobile app or computer system.
3. **Database Integration**
   Connecting the system to an external database or cloud server for automatic book data updates.
4. **Vision-Based Book Identification**
   Using a camera and image processing (OCR) to identify book titles without RFID cards.
5. **Autonomous Mapping**
   Implementing SLAM techniques for flexible navigation without predefined line paths.
6. **Multi-Robot Coordination**
   Deploying multiple robots working together for faster book retrieval in large libraries.
7. **Automated Charging System**
   Introducing a docking station for automatic battery charging.
8. **Improved Mechanical Design**
   Enhancing the robotic arm to handle books of different sizes and weights.
9. **Voice-Based Interaction**
   Allowing users to select books using voice commands for better accessibility.
10. **Integration with Library Management Systems**
    Connecting the robot with existing library ERP or management software.

5.5 CONCLUSION AND APPLICATIONS

Conclusion

The Automatic Book Picking Robot developed in this project demonstrates an effective and practical approach to automating book retrieval in library environments. By integrating an Arduino MEGA 2560 microcontroller with IR line follower sensors, RFID technology, DC motors, and a multi-servo robotic arm, the system successfully performs the complete operation of selecting, locating, identifying, and retrieving books with minimal human intervention.

The robot allows users to select a book by name using a simple button-based interface, while a 16×2 LCD display provides real-time system status and feedback. Navigation is achieved through reliable line-following logic, enabling accurate rack identification. The use of RFID technology ensures precise book identification even when multiple books are placed on the same shelf. The six-servo robotic arm enables controlled and safe book picking and handling.

Experimental results show that the system performs reliably in a controlled environment and meets the objectives of reducing human effort, saving time, and improving accuracy. The use of low-cost and easily available components makes the system affordable and suitable for educational institutions. Although the current implementation has some limitations, it provides a strong foundation for future smart library automation solutions.

Overall, this project successfully demonstrates the practical application of embedded systems, robotics, and automation in library management and serves as an effective prototype for further research and development.

Applications of the Proposed System

1. **Educational Institution Libraries**
   Can be used in schools, colleges, and universities to automate book retrieval.
2. **Digital and Smart Libraries**
   Forms a base model for smart library systems with advanced automation features.
3. **Research Libraries**
   Helps in managing large collections with reduced manual effort.
4. **Public Libraries**
   Improves service efficiency and reduces workload for library staff.
5. **Book Storage and Archives**
   Suitable for organized storage environments where items are arranged systematically.
6. **Academic Demonstrations**
   Useful for robotics, embedded systems, and automation demonstrations.
7. **Prototype for Industrial Automation**
   Serves as a learning platform for warehouse and inventory automation systems.

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